DC Charging System (BEVC-DC001)-III

Introduction

The rapid expansion of electric vehicles (EVs) necessitates robust and standardized charging infrastructure. At the heart of this infrastructure lies sophisticated communication protocols that enable seamless interaction between the EV and the charging station. This lecture delves into the intricacies of the DC Charging System (BEVC-DC001), focusing specifically on the underlying communication mechanisms. Understanding how EVs and chargers "talk" to each other is crucial for ensuring interoperability, safety, and efficient energy transfer, paving the way for a reliable and accessible EV ecosystem.

We will explore the role of CAN communication, the structure of CAN database files, and the specific adaptations made within the DC001 standard, which is largely derived from the GB/T 27930 protocol. This includes examining the physical and application layers of communication, the definition of message IDs, and the various stages of a charging session, each with its unique set of messages. Ultimately, mastering these communication protocols is key to developing and maintaining effective EV charging solutions.

Core Concepts

Term	Definition	Significance
`CAN database file`	Text files containing structured information about CAN messages, including signals, their parameters, scaling factors, units, and offsets, provided by device manufacturers.	Essential for interpreting raw CAN data into meaningful engineering units, enabling different nodes to understand and act upon received messages.
`PDU (Protocol Data Unit)`	A structured format for CAN data frames, defining how message IDs and data are organized for transmission between devices.	Standardizes the communication structure, ensuring that messages are consistently formatted and understood by all compliant devices.

Term	Definition	Significance
`PGN (Parameter Group Number)`	An 8-bit identifier within a CAN message ID that groups related parameters or signals together.	Allows receiving devices to quickly identify the type and purpose of a message, facilitating efficient processing of charging-related information.
`SPN (Signal Parameter Number)`	A unique identifier for individual signals within a specific parameter group.	Enables precise identification and extraction of individual data points from a message, crucial for detailed control and monitoring.
`DC001`	A specific DC charging system standard, primarily for low-power applications (up to 10kW for 48V, 15kW for 72V systems), incorporating elements from GB/T 27930 and AIS-138.	Defines the technical requirements for DC charging in India, ensuring interoperability and safety for a range of light electric vehicles.
`AIS-138 Part 2`	An Indian automotive industry standard that incorporates and adapts aspects of international charging protocols like GB/T 27930 for local requirements.	Provides the regulatory framework and specific communication message definitions for EV charging systems in India, including DC001.
`GB/T 27930`	A Chinese national standard for conductive charging communication protocols for electric vehicles, widely adopted and referenced by other standards like DC001.	Forms the foundational communication protocol for DC001, defining physical, data link, and application layer rules for EV-charger interaction.
`Scaling Factor`	A multiplier defined in the CAN database file used to convert raw data values from a CAN message into actual engineering units.	Crucial for translating raw, often integer-based, data into real-world values (e.g., volts, amps, percentage) that controllers can use.

Term	Definition	Significance
`Offset`	A value added to the scaled raw data, also defined in the CAN database file, to obtain the final engineering unit.	Accounts for baseline shifts or zero-point adjustments in sensor readings, ensuring accurate interpretation of data.

Detailed Analysis

1. The Role and Structure of CAN Database Files

CAN (Controller Area Network) database files are fundamental to the communication within the DC001 charging system. These text files serve as a dictionary, containing all the necessary information to interpret the raw data transmitted over the CAN bus. Device manufacturers typically provide these files, ensuring that different components can understand each other.

A CAN database file defines various aspects of CAN messages and their embedded signals:
- **Message Name, Message ID, and DLC (Data Length Code)**: Identifies the message and its priority, and specifies the number of bytes in the data field.
- **Channel/Signal Name**: Labels individual pieces of information within a message.
- **Location (Start Bit) and Size**: Specifies where a signal begins within the message's data field and its length in bits.
- **Signal Type**: Defines whether the signal is an integer, floating-point, etc.
- **Byte Order (Little-endian or Big-endian)**: Determines how multi-byte data is ordered.
- **Scaling Factor and Units String**: Provides the multiplier and units (e.g., RPM, V, A) needed to convert raw data into engineering units.
- **Range (Minimum or Maximum)**: Defines the expected operational limits of a signal.
- **Offset**: A value added after scaling to adjust the signal's baseline.

Example: Extracting Real Data from Raw CAN Message
Consider a scenario where a CAN message carries a raw data value of `0x03E8` (1000 in decimal) for "speed". If the CAN database defines a scaling factor of `0.2` and an offset of `100` for this signal, with units in RPM:
- Raw Data: `1000`
- Scaling Factor: `0.2`
- Offset: `100`
- Actual Data = (Raw Data * Scaling Factor) + Offset

- Actual Data = (1000 * 0.2) + 100 = 200 + 100 = 300 RPM.
This process demonstrates how raw, unintelligible data is transformed into meaningful information for the controller.

2. Overview of the BEVC-DC001 Standard

The BEVC-DC001 standard outlines the specifications for DC charging systems, particularly for light electric vehicles.
- **Input**: It supports 3-phase AC input.
- **Output**: It can deliver up to 10 kW for 48V systems and 15 kW for 72V systems.
- **Communication Derivation**: The detailed communication protocol is primarily derived from GB/T 27930-2015 and incorporated into AIS-138 Part 2, Annexure G, released by the Automotive Research Association of India (ARAI). It also references IEC 61851-24B.
- **CAN Communication**: The standard exclusively uses CAN communication for data exchange.
- **Connector**: It specifies the GB/T 20234.3 connector with a standard 5-meter cable.

3. Communication Protocol: Physical and Application Layers

The communication in DC001, largely based on GB/T 27930, follows a layered approach:
- **Physical and Data Link Layers**: These layers are based on SAE J1939 or ISO 11898 standards.
- **Baud Rate**: 250 kbps.
- **CAN Identifier**: Uses CAN 2.0b, which is a 29-bit identifier, allowing for 2^29 distinct messages.
- **Impedance**: The CAN line must have an impedance of 120 ohms for proper signal integrity.
- **Application Layer**: Defined by GB/T 27930-2015, this layer specifies the representation of parameters and parameter group numbers (PGNs).
- **Participants**: Only two participants communicate at a time: the charger and the vehicle's Battery Management System (BMS).
- **Fixed Addresses**: To ensure unambiguous communication, fixed hexadecimal addresses are assigned: `56H` for the charger and `F4H` for the BMS.
- **Dedicated Channel**: This CAN communication is dedicated solely to the charging system and does not interfere with other in-vehicle CAN systems (e.g., powertrain CAN).
- **Protocol Data Unit (PDU)**: CAN data frames are transmitted in the form of PDUs, which define their structure.

4. Message ID Format and Data Interpretation

The 29-bit message ID in DC001 (following GB/T 27930) is structured as a Protocol Data Unit (PDU) to convey specific information:
- **Priority Field (3 bits)**: Determines the message's urgency on the bus.
- **Reserved Bit**: A bit reserved for future use.
- **Data Page Bit**: Indicates the data page being used.
- **Parameter Group Number (PGN) (8 bits)**: Identifies the group of parameters contained within the message. This is crucial for the receiving node to understand the message's context.
- **Destination Address (8 bits)**: Specifies the intended recipient (e.g., `56H` for charger, `F4H` for BMS).
- **Source Address (8 bits)**: Identifies the sender of the message.

This structured message ID allows controllers to quickly ascertain the message's purpose, origin, and destination. Within each parameter group, individual signals are identified by a unique Signal Parameter Number (SPN).

5. Charging Stages and Message Exchange

The BEVC-DC001 standard defines six distinct stages of charging, each involving a specific set of messages exchanged between the EVSE (charger) and the EV (BMS):
1. **Handshake Stage**: Initial communication to establish connection.
- Messages: Charger Handshake Message (CHM), Battery Handshake Message (BHM).
2. **Handshake Recognition Stage**: Confirmation of compatibility.
- Messages: Charger Recognition Message (CRM), Battery Recognition Message (BRM).
3. **Parameter Configuration Stage**: Exchange of charging parameters.
- Messages: Battery Charge Parameter (BCP), Charger Time Synchronization (CTS), Charger Max/Min Output Parameter (CML), Charger Output Ready (CRO), Vehicle Charge Ready (BRO).
4. **Charging Stage**: Active power transfer.
- Messages: Battery Charge Requirement (BCL), Charger Charge State (CCS), Battery Charge State 1 (BSM), Battery Charge State 2 (BCS).
5. **Charging Suspension Stage**: Pausing the charging process.
- Messages: Charger Stopping Command (CST), Vehicle Stopping Command (BST).
6. **End-of-Charging Stage**: Termination of charging and data logging.
- Messages: Charger Statistic Data (CSD), Vehicle Statistic Data (BSD), Vehicle Receiving Error, Charger Receiving Error.

It's important to note that while the fundamental structure remains consistent, message names and specific parameters might differ slightly between GB/T 27930

and AIS-138 Part 2. AIS-138 Part 2 also defines a "Communication Error Stage" for handling faults.

6. BEVC-DC001 Application Layer Enhancements

The DC001 standard introduces several specific changes and additions to the application layer derived from GB/T 27930-2015 to cater to local requirements and enhance functionality:
- **Protocol Version Number**: Exchanged in the Charger Handshake Message (CHM).
- **Battery and Vehicle Information**: Includes battery type, manufacturer name, pack number, product code, charging times, property rights, and Vehicle Identification Number (VIN) in recognition messages.
- **Charger and Location Data**: Charger number, charging station location code, and recognition results in the Charger Recognition Message.
- **Charging Session Statistics**: Information like last charging end reason, status, start/end State of Charge (SoC), and distance traveled since the last charge.
- **Fault Codes**: A comprehensive list of fault codes for diagnostic purposes.

These additions provide more detailed data for monitoring, billing, and system diagnostics, crucial for a robust charging ecosystem.

7. Practical Example: Interpreting Vehicle Statistic Data Message

Let's trace the interpretation of a `Vehicle Statistic Data` message (message code BSD, PGN `001C00H`, priority 6, data length 7 bytes) sent from the BMS (source `F4H`) to the charger (destination `56H`).

Assume a received message ID `0x181C56F4` and raw data `46 012C 01A4 0A 37`.
1. **Message ID Decoding**:
- The 29-bit message ID `0x181C56F4` is parsed.
- The PGN `1C` (from `001C00H`) identifies it as a `Vehicle Statistic Data` message.
- The destination address `56H` confirms it's for the charger.
- The source address `F4H` confirms it's from the BMS.
2. **Raw Data Extraction**: The 7 bytes of data are `46H`, `012CH`, `01A4H`, `0AH`, `37H`.
- Byte 1: `46H` (70 decimal) for Final SOC.
- Bytes 2-3: `012CH` (300 decimal) for Minimum Cell Voltage.
- Bytes 4-5: `01A4H` (420 decimal) for Maximum Cell Voltage.
- Byte 6: `0AH` (10 decimal) for Minimum Battery Temperature.
- Byte 7: `37H` (55 decimal) for Maximum Battery Temperature.
3. **Actual Data Calculation (using resolution and offset from CAN database)**:
- **Final SOC**: Raw `70`. Resolution `1% / bit`. Offset `0`. Actual: `70 * 1 + 0 = 70%`.

- **Min Cell Voltage**: Raw `300`. Resolution `0.01 V / bit`. Offset `0`. Actual: `300 * 0.01 + 0 = 3.00 V`.
- **Max Cell Voltage**: Raw `420`. Resolution `0.01 V / bit`. Offset `0`. Actual: `420 * 0.01 + 0 = 4.20 V`.
- **Min Battery Temp**: Raw `10`. Resolution `1 deg C / bit`. Offset `0`. Actual: `10 * 1 + 0 = 10 deg C`.
- **Max Battery Temp**: Raw `55`. Resolution `1 deg C / bit`. Offset `0`. Actual: `55 * 1 + 0 = 55 deg C`.

This detailed example illustrates the complete process from receiving a raw CAN message to extracting and interpreting the actual, meaningful data that can be used by the charger controller or logged for analysis.

Key Takeaways

- [ ] CAN database files are essential for interpreting raw CAN bus data into meaningful engineering units.
- [ ] The BEVC-DC001 standard leverages CAN communication, primarily derived from GB/T 27930 and integrated into AIS-138 Part 2.
- [ ] Communication occurs between only two participants (charger and BMS) with fixed, unique hexadecimal addresses.
- [ ] The 29-bit CAN message ID follows a Protocol Data Unit (PDU) format, including priority, PGN, destination, and source addresses.
- [ ] Parameter Group Numbers (PGNs) allow for efficient identification of message types and the parameters they contain.
- [ ] Charging proceeds through six defined stages, each requiring specific message exchanges for proper operation.
- [ ] The DC001 standard includes additional parameters beyond GB/T 27930 for enhanced data collection and diagnostics.
- [ ] Raw data from CAN messages must be converted using scaling factors and offsets defined in the CAN database to obtain actual physical values.

Conclusion

The BEVC-DC001 standard, with its reliance on robust CAN communication and structured data interpretation, forms the backbone of efficient and interoperable DC charging for electric vehicles. By standardizing message formats, defining clear communication stages, and providing detailed parameter definitions through CAN database files, it ensures that chargers and EVs can seamlessly interact. This intricate dance of data exchange, from initial handshake to the final logging of charging statistics, is critical for the safety, reliability, and user experience of EV

charging infrastructure. As the EV landscape continues to evolve, understanding these foundational communication protocols will remain paramount for engineers, developers, and policymakers alike, driving the future of sustainable transportation.